



one version after another, improving the design each time. Their approach was simple: do not just discuss—keep building. Brainstorming helped bring in new ideas, but they believed real progress came from testing those ideas with quick prototypes. Government funding supported their prototyping efforts, so they did not face any major money issues while building and testing.

When discussing the electronics angle, Yatin added, “Electronics act as the engine of the system. All motor control and sensing rely on it. The electronics bring together all the mechanical parts, as motors are controlled through them. In total, the system uses eight different types of PCBs, each handling specific tasks like sensing, actuation, motor control, and processing. All these parts work together to make the system function, making electronics a core part of the product.”

The cooking robot has many sensors to help it work. A camera looks at the pan, and a thermal sensor checks the heat across it. A torque sensor tracks how much force is used

while stirring to understand how thick the food is. Position sensors like photo interrupters and encoders keep track of moving parts. There are also sensors to measure water and oil levels. For safety, anti-pinch sensors stop the motor if something blocks the way, and fire sensors turn everything off if it gets too hot.

AI plays a key role in helping the cooking robot know when the food is properly cooked, even if the ingredients change each time. It can tell when onions turn golden brown or when curry releases oil, using these signs to decide the right moment to move to the next step.

Manufacturing is handled by contract manufacturers, but the company has its own in-house teams for manufacturing oversight, quality assurance, customer service, technical support, software, AI and machine learning, mechanical and electronics, and design verification.

Testing plays a key role in ensuring reliability. Every design change goes through 5000 cooking cycles, and additional tests include vibration testing for road and sea

shipping, drop tests, and environmental chamber tests to check performance under different conditions.

While talking about the revenue, Yatin said, “We have sold 600 units so far, out of which 130 units have been delivered. The remaining 470 units are in different stages—some in logistics, others in manufacturing or on the production line. So far, around ₹3 crore has been received from customers, although the revenue has not been officially booked yet.”

As the product, company, and category are all new, the startup’s main challenge is building market trust and encouraging adoption by helping users understand its benefits and integrate it into daily routines.

Looking forward, the startup is also working on future product versions, improving cost efficiency, and optimising the supply chain. By combining deep tech with everyday cooking needs, the team aims to change how people approach food at home. The journey ahead is about scaling thoughtfully, keeping reliability high, and ensuring no one sleeps hungry.

—Nidhi Agarwal

3 AI-BASED QUALITY INSPECTION SYSTEMS BY KERALA STARTUP VERIFYGN

A Kerala startup deploys AI-driven hardware-software solutions to automate quality inspection in manufacturing.

VerifygnTech Private Limited, a Kerala-based deep tech startup founded in 2019, specialises in leveraging advanced AI-driven software to automate quality inspection processes for manufacturing and related industries. The company offers a hybrid of hardware and software products designed to be integrated into customers’ manufacturing lines.

Kesava Prasad T D, the Founder

and Director of VerifygnTech, elaborates on the scope of defect detection, stating, “Anything visible to a camera can be identified by our system.” Citing an example, he highlights the PCB industry, where their AI-powered solution is used to detect bent pins that may interfere with the assembly process—just one of many applications of their technology.

The startup has also developed systems for inspecting relay

modules, capable of detecting issues like improper wire crimping, loose connections, and incorrect wire colour sequences. More broadly, their advanced camera-based vision technology can identify a wide range of visually detectable defects. In PCB assembly, the system ensures correct component placement. It detects errors such as inserting the wrong part (like a capacitor instead of a resistor) or using components with incorrect specifications.

They aim to ensure defective parts—referred to as ‘NG’ or ‘not good’ in manufacturing terminology—are